



GRADING PIPELINE

Cost Estimation

Per-Call Pricing | Monthly Projections | Optimization Strategy

This document estimates the real-world cost of running the Wankr annotation rotation grading pipeline across all three AI providers. Costs are calculated using current published API pricing as of February 2026.

CURRENT API PRICING

All prices per 1 million tokens (1M tokens ≈ 750,000 words). Prices in USD.

PROVIDER	MODEL	TIER	INPUT / 1M	OUTPUT / 1M
Anthropic	Claude Haiku 4.5	Budget	\$1.00	\$5.00
Anthropic	Claude Sonnet 4.5	Mid-Tier	\$3.00	\$15.00
Anthropic	Claude Opus 4.5	Premium	\$5.00	\$25.00
OpenAI	GPT-4o-mini	Budget	\$0.15	\$0.60
OpenAI	GPT-4o	Mid-Tier	\$2.50	\$10.00
xAI	Grok 4.1 Fast	Budget	\$0.20	\$0.50
xAI	Grok 4	Mid-Tier	\$3.00	\$15.00

Cost Reduction Options

METHOD	SAVINGS	APPLICABLE?
Prompt Caching (Anthropic)	90% on repeated input	YES - grading docs are the same every call
Batch API (Anthropic/OpenAI)	50% on all tokens	YES - grading is not time-sensitive
Data Sharing (xAI)	\$150/mo free credits	MAYBE - shares prompts with xAI for training
Prompt Caching (xAI)	75% on cached input	YES - automatic, no config needed

TOKEN ESTIMATION PER CALL

Each grading call sends the annotator a set of context documents plus the response to grade, and receives back a structured grading output. Here is the token breakdown:

COMPONENT	TOKENS	NOTES
System prompt + grading instructions	~2,000	Role definition, grading criteria, output format
Context docs (persona rules, weights)	~4,000	Persona influences, focus/integrity rules, scenario matrix
Wankr response being graded	~800	The actual output under evaluation
Previous annotator's grading (rounds 2+)	~600	Only in rounds 2, 3, 4 of rotation
TOTAL INPUT per call	~7,400	Round 1 calls: ~6,800 (no previous grading)
Annotator grading output	~1,200	Structured scores, flags, explanations, evidence
TOTAL OUTPUT per call	~1,200	Consistent across all rounds

Pipeline Call Count

CONFIG	ROUNDS	CALLS PER ROUND	TOTAL CALLS	TOTAL INPUT TOKENS	TOTAL OUTPUT TOKENS
3 Annotators	3	3	9	~66,600	~10,800
4 Annotators	4	4	16	~118,400	~19,200

In the rotation system, each annotator grades in each round. Total input grows because later rounds include previous annotations as context. Total output stays consistent since every call produces the same structured grading format.

COST PER SINGLE API CALL

Using ~7,400 input tokens and ~1,200 output tokens per call. This is the cost of **one annotator grading one response in one round**.

PROVIDER	MODEL	INPUT COST	OUTPUT COST	TOTAL PER CALL
Anthropic	Haiku 4.5	$7.4K \times \$1/M = \0.0074	$1.2K \times \$5/M = \0.006	\$0.0134
OpenAI	GPT-4o-mini	$7.4K \times \$0.15/M = \0.0011	$1.2K \times \$0.60/M = \0.0007	\$0.0018
xAI	Grok 4.1 Fast	$7.4K \times \$0.20/M = \0.0015	$1.2K \times \$0.50/M = \0.0006	\$0.0021
Anthropic	Sonnet 4.5	$7.4K \times \$3/M = \0.0222	$1.2K \times \$15/M = \0.018	\$0.0402
OpenAI	GPT-4o	$7.4K \times \$2.5/M = \0.0185	$1.2K \times \$10/M = \0.012	\$0.0305
xAI	Grok 4	$7.4K \times \$3/M = \0.0222	$1.2K \times \$15/M = \0.018	\$0.0402
Anthropic	Opus 4.5	$7.4K \times \$5/M = \0.037	$1.2K \times \$25/M = \0.030	\$0.067

Key Takeaway

A single API call costs between **\$0.002** (GPT-4o-mini) and **\$0.067** (Opus 4.5). That is between one-fifth of a cent and seven cents per call. The cost adds up based on how many calls are in the full pipeline.

COST PER GRADED RESPONSE

This is the total cost to run **one Wankr response** through the complete annotation rotation pipeline. All annotators, all rounds.

3-Annotator Pipeline (9 calls per response)

Annotators: Claude + GPT + Grok. Each reviews the other two annotators' work across 3 rounds.

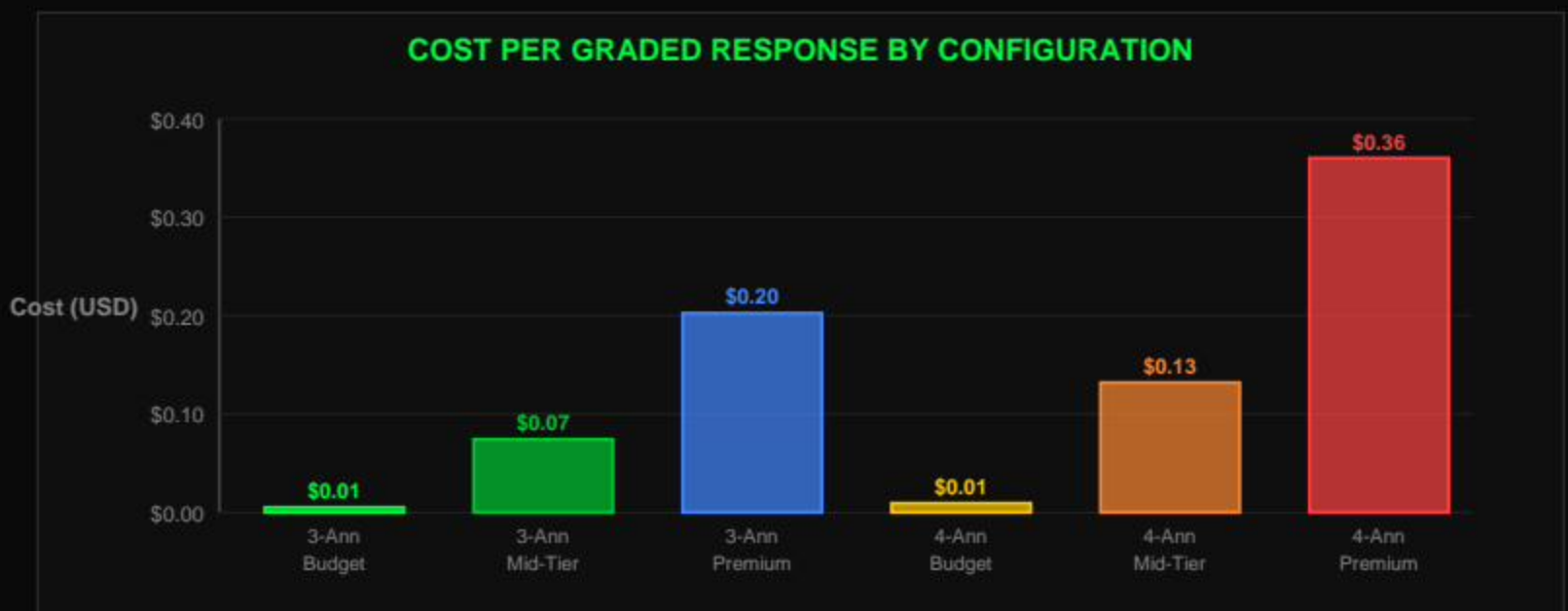
TIER	MODELS USED	9 CALLS TOTAL	EQUIVALENT
Budget	Haiku 4.5 + GPT-4o-mini + Grok 4.1 Fast	\$0.052	~5 cents per response
Mid-Tier	Sonnet 4.5 + GPT-4o + Grok 4	\$0.333	~33 cents per response
Premium	Opus 4.5 + GPT-4o + Grok 4	\$0.534	~53 cents per response

4-Annotator Pipeline (16 calls per response)

Annotators: Claude + GPT + Grok + User/Manual. 16 calls total (User rounds are free).

TIER	MODELS USED	16 CALLS TOTAL*	EQUIVALENT
Budget	Haiku 4.5 + GPT-4o-mini + Grok 4.1 Fast + User	\$0.069	~7 cents per response
Mid-Tier	Sonnet 4.5 + GPT-4o + Grok 4 + User	\$0.444	~44 cents per response
Premium	Opus 4.5 + GPT-4o + Grok 4 + User	\$0.712	~71 cents per response

*User/Manual rounds are free - only 12 of 16 calls are API calls. User annotator contributes grading at zero token cost.



MONTHLY COST PROJECTIONS

Based on your prototyping budget of \$20-50/month and manual trigger control:

Mid-Tier Models (Recommended for Quality)

RESPONSES/MO	3-ANN MID (\$0.33/ea)	4-ANN MID (\$0.44/ea)	TRAINING DATA POINTS
10	\$3.33	\$4.44	90 - 160 layers
25	\$8.33	\$11.10	225 - 400 layers
50	\$16.65	\$22.20	450 - 800 layers
100	\$33.30	\$44.40	900 - 1,600 layers
150	\$49.95	\$66.60	1,350 - 2,400 layers
200	\$66.60	\$88.80	1,800 - 3,200 layers

At \$20/month: ~60 responses with 3-annotator mid-tier, or ~45 with 4-annotator.

At \$50/month: ~150 responses with 3-annotator mid-tier, or ~113 with 4-annotator.



Budget Tier (Maximum Volume)

RESPONSES/MO	3-ANN BUDGET (\$0.05/ea)	4-ANN BUDGET (\$0.07/ea)	TRAINING DATA POINTS
100	\$5.20	\$6.90	900 - 1,600 layers
200	\$10.40	\$13.80	1,800 - 3,200 layers
500	\$26.00	\$34.50	4,500 - 8,000 layers
1000	\$52.00	\$69.00	9,000 - 16,000 layers

Budget models let you **grade 500+ responses for under \$35/month**. Quality trade-off: Haiku/mini/Fast are less capable annotators, so grading depth may be shallower.

COST OPTIMIZATION STRATEGY

1. Prompt Caching (Up to 90% Input Savings)

The grading context docs (persona rules, grading criteria, scenario matrix) are **identical for every call**. This is ~4,000 tokens that can be cached.

PROVIDER	CACHE BEHAVIOR	SAVINGS ON CACHED PORTION
Anthropic	5-min TTL cache, write 1.25x, read 0.1x	90% on reads after first write
OpenAI	Automatic prefix caching	50% on cached input
xAI	Automatic, no config needed	75% on cached input

Since grading runs process multiple calls rapidly, the cache stays hot. Estimated real-world savings: **40-60% reduction on input costs**.

2. Batch API (50% on Everything)

Grading is not time-sensitive - you do not need results in seconds. Both Anthropic and OpenAI offer batch processing with a **flat 50% discount** on all tokens. Trade-off: results come within 24 hours instead of immediately.

	STANDARD	BATCH (50% OFF)
Sonnet 4.5	\$3 / \$15	\$1.50 / \$7.50
GPT-4o	\$2.50 / \$10	\$1.25 / \$5.00

3. Hybrid Tier Strategy

Use different model tiers for different purposes:

- **Round 1 (initial grading):** Use mid-tier models - this is where grading quality matters most. The first pass sets the baseline.
- **Rounds 2-4 (review of reviews):** Use budget models - reviewing someone else's grading is a simpler task than doing the initial analysis.
- **Estimated savings:** 30-40% vs all mid-tier, with minimal quality loss on review rounds.

4. Combined Optimization Impact

OPTIMIZATION	SAVINGS	3-ANN MID COST/RESPONSE	4-ANN MID COST/RESPONSE
None (standard)	0%	\$0.33	\$0.44
+ Caching only	~40% input	\$0.24	\$0.32
+ Batch only	50% all	\$0.17	\$0.22
+ Cache + Batch	~60-65%	\$0.12	\$0.16

OPTIMIZATION	SAVINGS	3-ANN MID COST/RESPONSE	4-ANN MID COST/RESPONSE
+ Cache + Batch + Hybrid	~70-75%	\$0.08	\$0.11

With all optimizations: **grade ~600 responses/month for \$50**. That is **5,400 training data layers** from 3-annotator or **9,600 layers** from 4-annotator. More than enough to build a serious training corpus.

RECOMMENDED CONFIGURATION

Phase 1: Prototyping (Now)

SETTING	VALUE	REASONING
Annotators	3 (Claude + GPT + Grok)	Faster iteration, lower cost, 9 layers per response
Model Tier	Mid-Tier (Sonnet + 4o + Grok 4)	Quality grading needed to validate pipeline correctness
Volume	20-50 responses/month	Manual trigger, selective grading of best/worst responses
Budget	\$7-17/month	Well within \$20-50 limit
Optimizations	Caching ON, Batch OFF	Need fast feedback during prototyping

Phase 2: Scaling (After Pipeline Validated)

SETTING	VALUE	REASONING
Annotators	4 (Claude + GPT + Grok + User)	Maximum data quality with human oversight
Model Tier	Hybrid (mid for R1, budget for R2-4)	Quality where it matters, efficiency where it doesn't
Volume	100-200 responses/month	Batch processing, systematic coverage
Budget	\$11-22/month	Hybrid + caching + batch keeps cost low
Optimizations	Caching + Batch + Hybrid tiers	Maximum efficiency

Phase 3: Production (Training Corpus Ready)

SETTING	VALUE	REASONING
Annotators	4 (all AI + User spot-checks)	User reviews flagged disagreements only
Model Tier	Mid-tier all rounds	Consistent quality for final training data
Volume	500+ responses/month	Automated pipeline with quality gates
Budget	\$40-65/month	Cache + Batch keeps flagship models affordable
Training yield	8,000+ layers/month	Serious fine-tuning corpus in weeks

Bottom line: a single grading call costs \$0.002 to \$0.07 depending on model. A full pipeline run costs \$0.05 to \$0.71 per response. At your budget, you can grade 50-600 responses/month depending on tier and optimizations. This is extremely affordable for the training data quality it produces.